PUBLIC CANTEEN MANAGEMENT SYSTEM

Object Oriented Programming using Java [CS23333]

Database Management Systems [CS23332]

Submitted by

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of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

RAJALAKSHMI ENGINEERING COLLEGE, THANDALAM

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RAJALAKSHMI ENGINEERING COLLEGE

BONAFIDE CERTIFICATE

Certified that this project report titled **"PUBLIC CANTEEN MANAGEMENT SYSTEM"** is the bonafide work of **RISHI K (251001211)** and **SURYA PRAKASH C (251001279)**, who carried out the project work under my supervision.

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STUDENT DECLARATION

We hereby declare that the mini project entitled "PUBLIC CANTEEN MANAGEMENT SYSTEM" is carried out by our team as part of the academic requirements for the courses Object-Oriented Programming (CS23333) and Database Management Systems (CS23332) in the Bachelor of Technology programme in Information Technology at Rajalakshmi Engineering College.

We declare that this project report represents our work and understanding of Java programming, database management and JDBC connectivity.

We have referred to technical documentation and learning materials during the development of this project. The sources used are acknowledged in the References section.

We also declare that the implementation details and test results presented in this report will be verified against the actual project developed by our team.

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ACKNOWLEDGEMENT

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Finally, we thank the developers of Java, MySQL and JDBC documentation, which helped us understand the technologies used in this project.

ABSTRACT

The Canteen Management System is a command-line-based application developed using Java, MySQL and JDBC. The main objective of this project is to simplify the management of food items, customer details, orders and billing operations in a canteen.

In a traditional canteen system, maintaining food details, stock information and customer orders manually can be time-consuming. Manual record maintenance may also lead to calculation errors and difficulties in retrieving previous transactions.

The proposed Canteen Management System provides a computerised approach to managing these activities. The administrator can log in to the system, add new food items, view the menu, search for food items, update food details and delete records.

The system also supports customer information entry, order placement, bill calculation, stock quantity updates, order history and basic sales reporting.

Java is used to develop the application logic, while MySQL is used to store and manage the data. JDBC acts as the communication interface between Java and MySQL.

This project helps demonstrate important concepts of Object-Oriented Programming, Database Management Systems, SQL queries, CRUD operations and database connectivity.

Keywords: Java, MySQL, JDBC, Canteen Management, Database Management System, CRUD Operations.

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CHAPTER 1: INTRODUCTION

1.1 PROBLEM STATEMENT

Canteens handle a large number of food items and customer orders every day. Managing food details, stock quantities, customer information and billing manually can be difficult and time-consuming.

Maintaining records manually may result in calculation errors, incorrect stock information and difficulties in retrieving previous order details.

A computerised system is required to simplify these operations and maintain information in an organised manner.

The Canteen Management System is developed to reduce manual work and provide an efficient method for managing canteen activities using Java and MySQL.

1.2 PROJECT OVERVIEW

The Canteen Management System is a Java-based application that operates through the Command Prompt.

The system uses MySQL as the database and JDBC for communication between the Java application and the database.

The administrator can manage food items, maintain customer information, place orders, calculate bills and view previous order records.

The system stores information systematically and allows the administrator to retrieve and update records whenever required.

1.3 OBJECTIVES

The main objectives of this project are:

- To develop a simple canteen management application.
- To provide administrator login functionality.
- To manage food item details.
- To perform CRUD operations.
- To store customer information.
- To process customer orders.
- To calculate bills automatically.
- To update stock quantities after successful orders.
- To maintain order history.
- To generate basic sales reports.
- To demonstrate Java and DBMS concepts.

CHAPTER 1: INTRODUCTION (CONTD.)

1.4 SCOPE OF THE PROJECT

The scope of the Canteen Management System is to automate basic canteen operations through a command-line application.

The system focuses on managing food details, customer information, orders and billing operations.

The administrator can add, view, search, update and delete food items. Customer orders can be recorded and bills can be calculated based on the selected food items and quantities.

The system stores all important information in a MySQL database.

This project is designed for a small-scale canteen environment and can be enhanced with additional features in the future.

1.5 EXISTING SYSTEM

In the existing manual system, food details, customer orders and billing information are maintained using registers or handwritten records.

This process requires more manual effort and may result in errors while calculating bills or updating stock quantities.

Disadvantages of the Existing System:

- More manual work.
- Difficult to maintain large amounts of data.
- Possibility of calculation errors.
- Difficult to retrieve previous orders.
- Time-consuming stock management.
- More paperwork.

PROPOSED SYSTEM

The proposed Canteen Management System uses Java and MySQL to automate basic canteen operations.

It provides a menu-driven interface through which the administrator can manage food items and customer orders.

Advantages of the Proposed System:

- Reduces manual work.
- Provides organised data storage.
- Supports CRUD operations.
- Simplifies billing calculations.
- Maintains order history.
- Updates stock quantities.
- Provides faster data retrieval.

CHAPTER 2: REQUIREMENTS AND FEASIBILITY

2.1 HARDWARE REQUIREMENTS

The following hardware components are required to develop and execute the project.

Component	Requirement
Processor	Intel or equivalent processor
RAM	4 GB or above recommended
Hard Disk	Sufficient storage for Java and MySQL
Monitor	Standard display
Keyboard	Standard keyboard
Mouse	Standard mouse

2.2 SOFTWARE REQUIREMENTS

Software	Purpose
Operating System	Windows
Java JDK	Java program compilation and execution
MySQL	Database management
JDBC Driver	Database connectivity
Command Prompt	Running the Java application
Text Editor / IDE	Writing Java source code

2.3 FUNCTIONAL REQUIREMENTS

The system provides the following functionalities:

1. Administrator login.
2. Add food items.
3. View food menu.
4. Search food items.
5. Update food details.

6. Delete food items.
7. Store customer information.
8. Place customer orders.
9. Calculate bills.
10. Update stock quantities.
11. View order history.
12. Generate sales reports.

CHAPTER 2: REQUIREMENTS AND FEASIBILITY (CONTD.)

2.4 NON-FUNCTIONAL REQUIREMENTS

Usability: The system provides a simple command-line menu that allows the administrator to select the required operation.

Reliability: The application stores information in a MySQL database and retrieves the stored information when required.

Performance: The system performs basic database operations efficiently for a small-scale canteen application.

Maintainability: The application is divided into separate methods for different operations, making the program easier to understand and maintain.

Data Integrity: The database uses primary keys and foreign keys to maintain relationships between records.

2.5 FEASIBILITY STUDY

Technical Feasibility: The project uses Java, MySQL and JDBC, which are suitable technologies for developing a database-driven application.

Economic Feasibility: The project can be developed using commonly available Java and MySQL software without requiring expensive infrastructure.

Operational Feasibility: The system is simple to operate through the Command Prompt. An administrator with basic computer knowledge can use the application.

Schedule Feasibility: The project can be completed through different stages such as requirement analysis, database design, coding, testing and documentation.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 SYSTEM ARCHITECTURE

The Canteen Management System follows a simple application-to-database architecture.

The administrator interacts with the Java application through the Command Prompt.

The Java application processes the input and communicates with the MySQL database using JDBC.

The database stores the food details, customer information and order records.

ARCHITECTURE FLOW

```
ADMINISTRATOR ↓ COMMAND PROMPT ↓ JAVA APPLICATION ↓ JDBC CONNECTIVITY ↓
MYSQL DATABASE
```

Diagram Explanation: The administrator enters the required information through CMD. The Java program processes the input and sends database queries through JDBC. MySQL executes the queries and returns the results to the Java application. The results are displayed to the administrator.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN (CONTD.)

3.2 MODULE DESCRIPTION

The Canteen Management System consists of the following modules:

1. Administrator Login Module: This module allows the administrator to access the application using a username and password. The system verifies the entered login credentials before displaying the main menu.

2. Food Management Module: This module manages food item details such as food ID, food name, price and available quantity. The administrator can add, view, search, update and delete food records.

3. Customer Management Module: This module stores customer information such as customer name and phone number. The customer details are associated with the orders placed in the canteen.

4. Order Management Module: This module allows the administrator to enter customer orders by selecting food items and quantities. The order information is stored in the database.

5. Billing Module: This module calculates the subtotal for each food item and determines the total bill amount.

6. Stock Management Module: When an order is placed successfully, the available quantity of the selected food items is updated.

7. Order History and Sales Report Module: This module retrieves previous order details and displays basic sales information.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN (CONTD.)

3.3 SYSTEM FLOW DIAGRAM

The system flow diagram represents the sequence of operations performed in the Canteen Management System.

```
START ↓ ADMIN LOGIN ↓ VALIDATE LOGIN DETAILS ↓ IS LOGIN VALID? ├── If NO → Display Error Message → Login Again. └── If YES → Display Main Menu. ↓ SELECT OPERATION 1. Add Food 2. View Food 3. Search Food 4. Update Food 5. Delete Food 6. Place Order 7. Order History 8. Sales Report 9. Exit ↓ PERFORM SELECTED OPERATION ↓ DISPLAY RESULT ↓ RETURN TO MAIN MENU ↓ EXIT → STOP
```

Explanation: The application starts with administrator login. After successful authentication, the main menu is displayed. The administrator selects an operation and the system performs the corresponding database activity. The result is displayed and the application returns to the main menu until the administrator selects Exit.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN (CONTD.)

3.4 DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram represents the movement of information between the user, application and database.

Level 0 DFD Components:

- **External Entity:** Administrator
- **Process:** Canteen Management System
- **Data Store:** MySQL Database

DATA FLOW SEQUENCE

```
Administrator ↓ [Login Details / Food Details / Order Details] Canteen Management System ↓ [Database Queries] MySQL Database ↓ [Stored Records / Query Results] Canteen Management System ↓ [Displayed Output] Administrator
```

Explanation: The administrator provides input to the Canteen Management System. The application processes the information and communicates with the MySQL database through JDBC. The database stores and retrieves the required records. The application displays the results to the administrator through the Command Prompt.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN (CONTD.)

3.5 USE CASE DIAGRAM

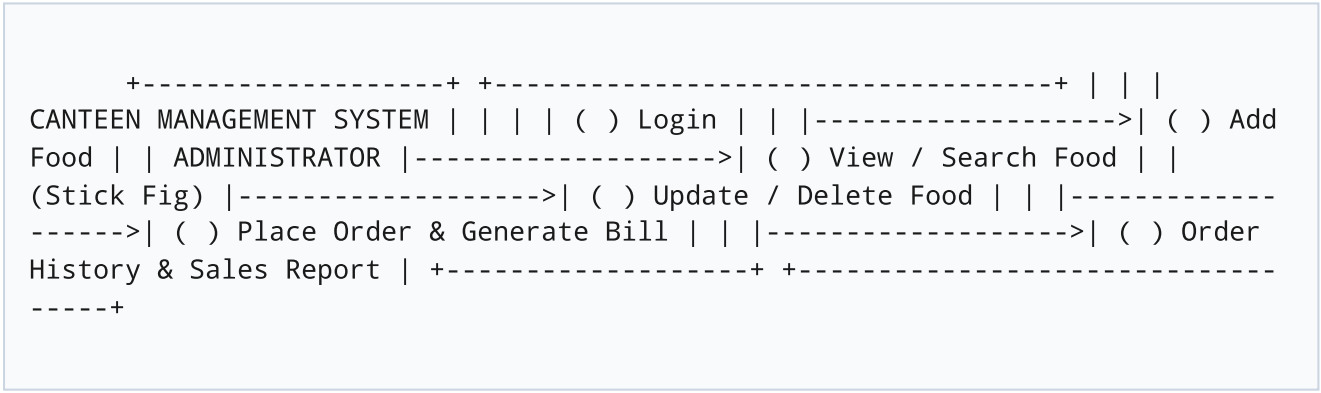
A Use Case Diagram represents the interaction between the administrator and the Canteen Management System.

Actor: Administrator

Use Cases:

- Login
- Add Food
- View Food
- Search Food
- Update Food
- Delete Food
- Manage Customer Details
- Place Order
- Generate Bill
- View Order History
- View Sales Report
- Logout / Exit

DIAGRAM REPRESENTATION INSTRUCTIONS



Explanation: The administrator is the main actor of the system. The administrator interacts with the application to manage food details, process customer orders and retrieve sales information.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN (CONTD.)

3.6 ENTITY RELATIONSHIP DIAGRAM (ER DIAGRAM)

The Entity Relationship Diagram represents the database entities and their relationships.

Entities and Attributes:

- **ADMIN:** Admin_ID, Username, Password
- **FOOD:** Food_ID, Food_Name, Price, Quantity
- **CUSTOMER:** Customer_ID, Customer_Name, Phone
- **ORDERS:** Order_ID, Customer_ID, Order_Date, Total, Payment_Mode
- **ORDER_DETAILS:** Detail_ID, Order_ID, Food_ID, Quantity, Subtotal

Relationships:

1. One customer can place many orders.
2. Each order belongs to one customer.
3. One order can contain multiple order-detail records.
4. One food item can appear in multiple order-detail records.

DIAGRAM INSTRUCTIONS

Draw five rectangles representing the five entities. Connect CUSTOMER to ORDERS, ORDERS to ORDER_DETAILS, and FOOD to ORDER_DETAILS. Mark the primary keys and foreign keys in the respective entities.

CHAPTER 4: DATABASE DESIGN

4.1 DATABASE OVERVIEW

The database used in this project is named **CanteenDB**. MySQL is used as the Relational Database Management System.

The database stores administrator information, food item details, customer information, orders and order details.

The database consists of five main tables:

1. Admin
2. Food
3. Customer
4. Orders
5. Order_Details

These tables are connected using primary keys and foreign keys.

4.2 TABLE STRUCTURE

ADMIN TABLE

Field Name	Data Type	Description
Admin_ID	INT	Primary Key
Username	VARCHAR(50)	Admin username
Password	VARCHAR(255)	Admin password

FOOD TABLE

Field Name	Data Type	Description
Food_ID	INT	Primary Key
Food_Name	VARCHAR(100)	Food name
Price	DECIMAL(10,2)	Food price
Quantity	INT	Available quantity

CHAPTER 4: DATABASE DESIGN (CONTD.)

CUSTOMER TABLE

Field Name	Data Type	Description
Customer_ID	INT	Primary Key
Customer_Name	VARCHAR(100)	Customer name
Phone	VARCHAR(20)	Phone number

ORDERS TABLE

Field Name	Data Type	Description
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Order_Date	DATETIME	Order date
Total	DECIMAL(10,2)	Total bill
Payment_Mode	VARCHAR(30)	Payment method

ORDER_DETAILS TABLE

Field Name	Data Type	Description
Detail_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Food_ID	INT	Foreign Key
Quantity	INT	Ordered quantity

Field Name	Data Type	Description
Subtotal	DECIMAL(10,2)	Item subtotal

4.3 PRIMARY KEY AND FOREIGN KEY

Primary Key: A Primary Key uniquely identifies each record in a database table. Examples: `Admin_ID`, `Food_ID`, `Customer_ID`, `Order_ID`, `Detail_ID`.

Foreign Key: A Foreign Key is used to establish a relationship between two tables. Examples: `Customer_ID` in Orders, `Order_ID` in Order_Details, `Food_ID` in Order_Details.

CHAPTER 4: DATABASE DESIGN (CONTD.)

4.4 NORMALISATION

Normalisation is the process of organising data in a database to reduce unnecessary duplication and improve data consistency.

The Canteen Management System separates administrator, food, customer, order and order-detail information into different tables.

First Normal Form (1NF)

Each table contains individual values in its columns. Repeating groups are avoided.

Second Normal Form (2NF)

Non-key attributes should depend on the complete primary key.

Third Normal Form (3NF)

Non-key attributes should depend only on the primary key and not on other non-key attributes.

The database structure uses separate tables for different entities and connects them through keys. This helps organise information and reduce repeated data.

CHAPTER 5: IMPLEMENTATION

5.1 DEVELOPMENT ENVIRONMENT

The Canteen Management System is developed using Java and MySQL.

Java is used for implementing the application logic and menu-driven operations. MySQL is used to store and manage the data. JDBC is used to establish communication between Java and MySQL.

The application is executed using the Windows Command Prompt.

Technologies Used:

- **Java:** Used to develop the application.
- **MySQL:** Used to store food, customer and order details.
- **JDBC:** Used to connect Java with the MySQL database.
- **CMD:** Used to compile and execute the Java program.

5.2 JDBC CONNECTIVITY

Java Database Connectivity is an API that allows Java applications to communicate with databases.

In this project, JDBC is used to connect the Java program to the MySQL database. The JDBC connection allows the application to execute SQL statements, retrieve records and modify database information.

JDBC Connection URL Format:

```
jdbc:mysql://localhost:3306/CanteenDB
```

The database username and password must match the local MySQL configuration.

CHAPTER 5: IMPLEMENTATION (CONTD.)

5.3 ADMINISTRATOR LOGIN

The administrator login module provides controlled access to the application.

The administrator enters a username and password. The application checks the entered credentials with the administrator information stored in the database.

If the credentials are valid, the main menu is displayed. If the credentials are invalid, an error message is displayed.

5.4 FOOD MANAGEMENT

The Food Management module is used to maintain the food items available in the canteen. The administrator can perform the following operations:

ADD FOOD: This operation allows the administrator to enter a new food item with its name, price and quantity. The details are stored in the Food table.

VIEW FOOD: This operation displays the available food items stored in the database.

SEARCH FOOD: This operation retrieves a particular food item using its ID or other supported search criteria.

UPDATE FOOD: This operation allows the administrator to modify existing food item details.

DELETE FOOD: This operation removes a selected food record from the database, subject to database constraints.

CHAPTER 5: IMPLEMENTATION (CONTD.)

5.5 ORDER MANAGEMENT

The Order Management module is used to record customer orders.

The administrator enters the customer details and selects the required food items and quantities. The application stores the customer information and creates an order record. The selected food items are stored in the Order_Details table.

BILLING

The billing module calculates the total amount based on the selected food items and quantities.

Billing Formula:

```
      Subtotal = Food Price × Ordered Quantity Total Bill = Sum of All Item
Subtotals
```

Illustrative Billing Example:

Food Item	Price	Quantity	Subtotal
Idli	₹10	2	₹20
Dosa	₹30	1	₹30
Tea	₹12	2	₹24
		Total	**₹74**

* The above values are only an illustrative billing example.

STOCK MANAGEMENT

After a successful order, the available quantity of the selected food items is reduced based on the ordered quantity. This helps maintain updated stock information in the database.

CHAPTER 5: IMPLEMENTATION (CONTD.)

5.6 ORDER HISTORY

The Order History module is used to retrieve previously recorded customer orders.

The application retrieves the order information stored in the MySQL database. This helps the administrator review previous transactions.

SALES REPORT

The Sales Report module displays basic sales information based on the stored order records. The report helps the administrator review the recorded order totals.

5.7 JAVA OOP CONCEPTS USED

Java supports Object-Oriented Programming concepts such as classes, objects, methods, encapsulation, inheritance and polymorphism. In this project, Java classes and methods are used to organise the application logic.

Class: A class is a blueprint used to define variables and methods.

Object: An object is an instance of a class.

Methods: Methods are used to perform specific operations such as adding food, viewing food and placing orders.

> *Important Note:* Include inheritance, polymorphism and encapsulation as implemented concepts only if your final source code actually demonstrates them. Do not claim concepts that are not present in your program.

CHAPTER 6: VERIFICATION AND TEST PLAN

6.1 TESTING

Testing is performed to verify whether the application works according to the expected requirements.

The Canteen Management System should be tested by performing each operation and checking the corresponding database records.

6.2 TEST CASES

Test ID	Test Scenario	Expected Result
TC01	Valid admin login	Login successful
TC02	Invalid admin login	Error message displayed
TC03	Add food item	Food record inserted
TC04	View food items	Food list displayed
TC05	Search food item	Matching record displayed
TC06	Update food details	Record updated
TC07	Delete food item	Record deleted or safely rejected if referenced
TC08	Place valid order	Order stored successfully
TC09	Calculate bill	Correct total displayed
TC10	View order history	Previous orders displayed
TC11	Generate sales report	Sales summary displayed

6.3 TESTING RESULT

Each test case should be executed using the actual application. The expected output should be compared with the actual output. Any errors identified during testing should be corrected before the final demonstration.

Note: Fill in the actual test results after running your Java program. Do not mark a test as successful unless it has been tested.

CHAPTER 7: SECURITY, LIMITATIONS & MAINTENANCE

7.1 SECURITY

Security is important for protecting the data stored in the Canteen Management System. The following security measures should be considered:

- Administrator login is provided to control access.
- User inputs should be validated.
- `PreparedStatement` should be used for SQL queries containing user input.
- Database credentials should not be shared publicly.
- Passwords should be securely hashed in a real-world application.
- Database transactions should be used for related order and stock operations.

7.2 LIMITATIONS

The current system has the following limitations:

- The application uses a Command Prompt interface.
- It does not provide a graphical user interface.
- It is designed mainly for a small-scale canteen.
- Online ordering is not supported.
- Online payment gateway integration is not included.
- Customer mobile application support is not available.
- Advanced inventory management is not included.

7.3 MAINTENANCE

The database should be backed up regularly to avoid data loss.

Food details and stock quantities should be updated whenever necessary.

The application should be checked periodically to identify errors.

Java and MySQL software should be maintained using compatible versions.

The database structure should be modified carefully to preserve existing records and relationships.

CHAPTER 8: CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

The Canteen Management System is developed to simplify basic canteen operations using Java, MySQL and JDBC.

The application provides features for managing food items, maintaining customer information, recording orders, calculating bills and retrieving order history.

The use of MySQL provides structured storage for the application data, while JDBC establishes communication between the Java program and the database.

This project demonstrates the practical application of Java programming, SQL queries, CRUD operations, database connectivity and relational database concepts.

Through this project, we gained practical knowledge of application development, database design and testing.

The Canteen Management System provides a basic foundation for managing canteen activities in an organised manner.

8.2 FUTURE ENHANCEMENTS

The following features can be added in the future:

- Graphical User Interface (GUI).
- Online food ordering facility.
- Customer login and registration.
- Online payment integration.
- Receipt printing facility.
- Advanced sales reports.
- Low-stock alerts.
- Supplier management.
- Multi-branch canteen support.
- Web or mobile application support.

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6. Course materials for Database Management Systems (CS23332), Rajalakshmi Engineering College.

APPENDIX A: SQL SCHEMA

The following SQL statements represent the proposed database structure for the Canteen Management System. Verify the column names and constraints against the actual database used in your project before submission.

```
CREATE DATABASE CanteenDB; USE CanteenDB; CREATE TABLE Admin ( Admin_ID INT PRIMARY KEY AUTO_INCREMENT, Username VARCHAR(50) NOT NULL UNIQUE, Password VARCHAR(255) NOT NULL ); CREATE TABLE Food ( Food_ID INT PRIMARY KEY AUTO_INCREMENT, Food_Name VARCHAR(100) NOT NULL, Price DECIMAL(10,2) NOT NULL, Quantity INT NOT NULL ); CREATE TABLE Customer ( Customer_ID INT PRIMARY KEY AUTO_INCREMENT, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHAR(20) ); CREATE TABLE Orders ( Order_ID INT PRIMARY KEY AUTO_INCREMENT, Customer_ID INT NOT NULL, Order_Date DATETIME DEFAULT CURRENT_TIMESTAMP, Total DECIMAL(10,2) NOT NULL, Payment_Mode VARCHAR(30), FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID) ); CREATE TABLE Order_Details ( Detail_ID INT PRIMARY KEY AUTO_INCREMENT, Order_ID INT NOT NULL, Food_ID INT NOT NULL, Quantity INT NOT NULL, Subtotal DECIMAL(10,2) NOT NULL, FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID), FOREIGN KEY (Food_ID) REFERENCES Food(Food_ID) );
```

APPENDIX B: OUTPUT SCREENSHOTS

The following console output representations depict the execution flow of the command-line interface developed in Java for the Canteen Management System.

B.1 Administrator Login

```
      ================================================= CANTEEN MANAGEMENT SYSTEM - LOGIN
================================================= Enter Admin Username : admin Enter Admin
Password : ******** [SUCCESS] Authentication successful. Welcome, Admin! --------------------
----------------------------
```

B.2 Main Menu

```
      ================================================= MAIN MENU
================================================= 1. Add Food Item 2. View Food Menu 3.
Search Food Item 4. Update Food Item 5. Delete Food Item 6. Place Order & Generate Bill 7.
View Order History 8. View Sales Report 9. Exit ----------------------------------------
---- Enter your choice (1-9):
```

B.3 Add Food Item

```
      --- ADD NEW FOOD ITEM --- Enter Food Name : Masala Dosa Enter Price (₹) : 45.00 Enter
Quantity : 50 [SUCCESS] Food Item 'Masala Dosa' added successfully with Food_ID: 104.
```

B.4 View Food Menu

```
      +---------+--------------------+------------+----------+ | Food_ID | Food Name | Price
(₹) | Quantity | +---------+--------------------+------------+----------+ | 101 | Idli (2 Pcs)
| 20.00 | 40 | | 102 | Plain Dosa | 35.00 | 30 | | 103 | Tea / Coffee | 15.00 | 100 | | 104 |
Masala Dosa | 45.00 | 50 | +---------+--------------------+------------+----------+
```

B.5 Search Food Item

```
      Enter Search Keyword or Food_ID: 102 [RESULT FOUND] Food ID: 102 | Name: Plain Dosa |
Price: ₹35.00 | Available Stock: 30
```

APPENDIX B: OUTPUT SCREENSHOTS (CONTD.)

B.6 Update Food Item & B.7 Delete Food Item

```
--- UPDATE FOOD ITEM --- Enter Food ID to Update : 101 Enter New Price (₹) : 25.00
[SUCCESS] Record for Food_ID 101 updated successfully. --- DELETE FOOD ITEM --- Enter Food ID
to Delete : 104 [SUCCESS] Record for Food_ID 104 deleted successfully.
```

B.8 Place Order & B.9 Bill Generation

```
--- PLACE CUSTOMER ORDER --- Enter Customer Name : Ramesh Kumar Enter Customer Phone :
9876543210 Enter Food ID : 101 | Quantity: 2 Enter Food ID : 103 | Quantity: 1
================================================== CANTEEN OFFICIAL RECEIPT
================================================== Customer: Ramesh Kumar Phone: 9876543210 --
-------------------------------------------- Item Price Qty Subtotal ---------------------
---------------------------- Idli (2 Pcs) 25.00 2 ₹50.00 Tea / Coffee 15.00 1 ₹15.00 --------
---------------------------------------- TOTAL AMOUNT DUE: ₹65.00 Payment Mode: Cash
[SUCCESS] Order #1001 created & stock updated automatically.
==================================================
```

B.10 Order History & B.11 Sales Report

```
+----------+--------------+--------------------+----------+--------------+ | Order_ID
| Customer Name | Order Date | Total(₹) | Payment Mode | +----------+--------------+---------
-----------+----------+--------------+ | 1001 | Ramesh Kumar | 2026-11-02 10:15:00 | 65.00 |
Cash | +----------+--------------+--------------------+----------+--------------+ TOTAL
SALES REVENUE RECORDED: ₹65.00 | TOTAL ORDERS: 1
```

END OF PROJECT REPORT